

Voice of the Courts: An Automated AI Legal Podcast and Digest

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The Challenge and The Proposed Approach

A core challenge in the Canadian legal landscape is that justice is often inaccessible not because it is secret, but because it is incomprehensible. The Access to Algorithmic Justice (A2AJ) project has made a monumental contribution by gathering over 116,000 decisions and 5,000 laws into open-access datasets. However, for the vast majority of Canadians, legal clinics, advocacy groups, and even journalists, this raw data remains an impenetrable wall of text. The sheer volume and dense "legalese" of new weekly decisions from the Supreme Court, Federal Court, and critical tribunals (like the Immigration and Refugee Board or the Social Security Tribunal) create a significant barrier. This information gap is a fundamental **access to justice challenge**: if communities cannot understand the legal system as it evolves, they cannot effectively navigate it, advocate for change, or understand their rights. This system will function as an "AI-powered legal journalist,"

automatically performing an end-to-end process that ingests raw legal decisions and publishes its analysis as clear, plain-language articles and audio podcast episodes on a free, open-access platform. This automated process includes the following key components:

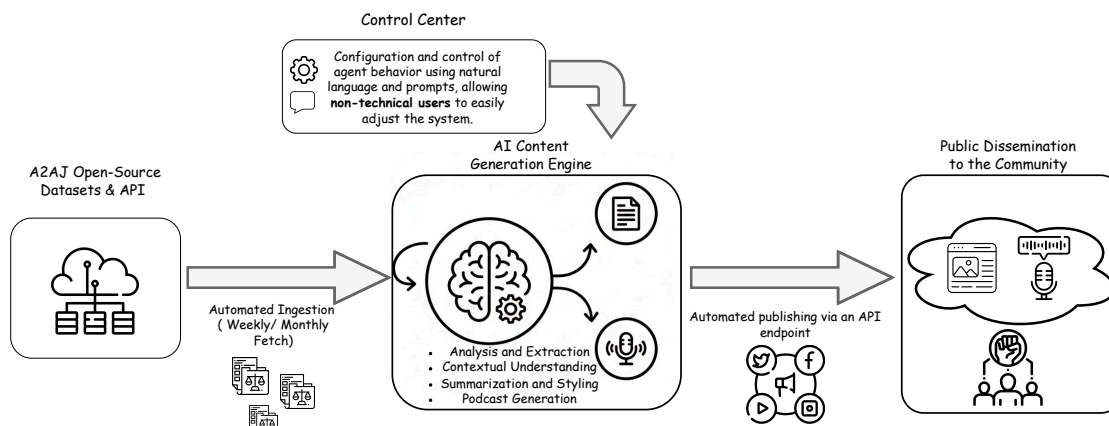


Figure 1: The system automates the entire process from data to public understanding: 1) Ingesting A2AJ datasets, 2) Analyzing and transforming them into summaries and podcasts via the AI Engine, and 3) Disseminating the content to the community. The Control Center is the key interface, enabling non-technical users to adjust the system's behavior using natural language.

1. **Ingest:** It will connect directly to the A2AJ API to monitor new decisions and legislation.
2. **Analyze and Synthesize:** The core of the system uses a configurable network of Large Language Model (LLM) agents to process the ingested data. Crucially, this system leverages intuitive, 'low-code' open-source frameworks (such as *LangGraph*, *Dify*, or *n8n*) to ensure its behavior is easily manageable by non-technical users, eliminating the need for complex coding to change the system's focus or tone.
3. **Control Center:** A simple, non-technical management interface that empowers any user or organization to customize the output to their specific needs. This ensures maximum accessibility and utility. Instead of a one-size-fits-all summary, organizations can use this "control center" to easily adjust key parameters such as:
 - **Audience & Tone:** Instantly change the output style (e.g., "Plain-language summary for the public," "Technical brief for a legal clinic," or "Informal podcast script").
 - **Level of Detail:** Define the desired length and granularity (e.g., "Single-paragraph key finding," "Top 5 takeaways," or "Comprehensive multi-point summary").
 - **Thematic Focus:** Direct the AI to analyze and report on specific aspects of a decision (e.g., "Focus only on the impact on refugees" or "Summarize changes to disability benefit rules").
4. **Publish:** This content will be published as accessible public media such as plain-language articles and podcast episodes on a free, open-access platform.
5. **Package & Distribute:** The entire end-to-end platform will be fully dockerized¹ and released as an

¹In non-technical terms, 'dockerizing' means packaging the entire software platform—including all its complex components,

open-source tool. This ensures any person or organization can deploy their own complete instance with minimal technical effort, allowing them to adapt the system for their specific community needs.

This project directly aligns with A2AJ's mandate and the "Example Projects" by **building an automated summarization tool** and **creating an automated generative AI podcast** using the A2AJ datasets. Furthermore, the entire system will be **dockerized and released as an easy-to-deploy open-source tool**, allowing other institutions to adapt it for their own needs.

Access to Justice Impact

This project is designed to generate **substantial, measurable impact** and **advance access to justice** across multiple fronts. This impact is amplified by the platform's core design: it is both **powerful and accessible**. The system is engineered for flexibility, allowing users with minimal technical skill to easily customize its behavior. This adaptability ensures any organization can tailor the tool for their specific purpose, whether serving a particular community or analyzing a unique set of legal data.

Direct Community Impact

This platform directly empowers individuals by translating dense legal text into plain-language audio. A new Canadian can listen to a 5-minute summary of key Immigration and Refugee Board decisions. A person with disabilities can stay informed about Social Security Tribunal trends without hiring a lawyer. The platform makes legal developments accessible to those with literacy challenges, visual impairments, or simply limited time.

Institutional Capacity Building

This tool acts as a force multiplier for frontline organizations that serve marginalized populations. Legal aid clinics, non-profits, and advocacy groups currently spend countless hours manually scanning for relevant decisions. Our tool provides them with an automated "weekly digest," freeing up their valuable, limited resources to focus on direct client representation and systemic advocacy.

Systemic Improvements & Increased Understanding

This project creates lasting systemic change by closing the critical "last mile" gap between open data and genuine public understanding. By automating this process, we create a scalable and sustainable way to dramatically **increase the public's understanding of access to justice issues**, demystifying the courts and **ultimately benefiting those facing legal barriers** by equipping them and their advocates with timely, understandable information.

Technical Innovation and A2AJ Integration

This project is not merely a *user* of the A2AJ datasets; it is a critical *enhancement* and *expansion* of the A2AJ infrastructure itself. It directly addresses the "last mile" problem of legal data by building an automated, configurable bridge from A2AJ's raw data archive to public understanding.

settings, and AI agents—into a single, self-contained 'kit.' This allows any person or organization to install and run the complete system on their own computer or server with a single command, without needing to be an IT expert or manually configure complex software. It makes advanced technology simple to deploy and use.

Transforming a Passive Archive into an Active Resource

The A2AJ datasets are a monumental achievement in data collection. This project provides the essential **downstream accessibility layer** that unlocks their potential.

- **Leveraging:** The system's "Ingest" component is designed to hook directly into the A2AJ API, treating it as a live, dynamic source. It leverages the A2AJ infrastructure as the foundational "fuel" for the entire platform.
- **Enhancing:** It enhances A2AJ's capabilities by adding a powerful, automated synthesis engine that transforms raw legal decisions into entirely new, high-value assets. This process significantly expands the *utility* and *modality* of the A2AJ data, turning a passive archive of text into an active, multi-format (text, audio) public resource. The system systematically creates plain-language summaries, structured thematic digests, and broadcast-quality audio podcasts where only dense legal text existed before.

Novel Solutions to Legal Information and Design Problems

The project's primary technical innovation is not just in applying AI, but in its **human-centric and deployable architecture**, which solves two critical problems in legal tech: complexity and accessibility.

- **The "Control Center" as a Design Innovation:** The core problem is that legal information needs are specific and varied. A one-size-fits-all summary is useless. The "Control Center" is a novel solution to this *design problem*. By using high-level frameworks (like *LangGraph* or *Dify*), it abstracts away the immense complexity of managing AI agents. It empowers a legal aid clinic manager—not an AI engineer—to retask the entire system with simple natural language (e.g., "Focus on refugee cases and make the tone supportive"). This **democratizes the power of generative AI** for the exact frontline organizations A2AJ aims to serve.
- **"Dockerization" as a Deployment Innovation:** Most academic tech projects fail at the point of adoption because they are too difficult to install and maintain. Our solution to this *technical problem* is to "dockerize" the entire platform. As detailed in our footnote, this packages the complete system (the AI agents, the Control Center, the web server) into a **single, easy-to-deploy "kit"**. This is a technical approach for this sector, ensuring that any organization, regardless of its IT budget, can launch its own private, customized instance of the platform with minimal set of commands. This design ensures our tool is not just a proposal but a scalable, distributable, and sustainable contribution to the A2AJ ecosystem.

Technical Feasibility and Team Capacity

The technical goals of this project are ambitious yet entirely feasible, as they align directly with the core expertise and proven experience of the project lead, Hassan Hamidi. The team's capacity to deliver this project is not hypothetical; it is based on successfully executing analogous systems in professional and applied research contexts:

- **Strong Foundational & Academic Grounding:** The proposed work is underpinned by a solid academic background, including a Bachelor's degree in computer engineering, a Master's degree in AI, and ongoing PhD research in Computer Science centered on generation AI models. This is further enhanced by expert-level skills in the essential project tools, such as **Python**, **PyTorch**, and the **Hugging Face** ecosystem (Transformers, TRL).

- **Demonstrated Proficiency in Applied LLM Legal-Tech:** The candidate possesses hands-on experience in creating similar systems. He crafted a "Legal Conflict Detection System" for the analysis of legal documents. Additionally, he has built an "Article Assessment System" in his role as an Applied ML Engineer. This collective experience attests to his capability to handle legal texts and develop the proposed configurable LLM service.

To summarize, the project leader has an excellent mix of skills, including academic expertise in AI and practical experience in creating and implementing similar LLM systems as suggested for legal-tech.

Project Plan & Timeline

We have structured this project over a 8-month timeline, organized into five key phases. This plan is designed to be achievable, with clear milestones to ensure accountability and progress. Each phase builds upon the last:

- **Phase 1: Data Investigation & Architectural Design (Months 1-2)** This foundational phase is dedicated to a deep analysis of the A2AJ datasets to understand their structure, metadata, content types, and update frequency. We will use these insights to finalize the system's high-level architecture, defining the precise data flow between the "Ingest," "Analyze," "Control," and "Publish" components. This phase also includes a critical evaluation and selection of the optimal open-source frameworks (e.g., *LangGraph*, *Dify*, or *n8n*), text-to-speech (TTS) engines, and deployment platforms (e.g., Docker) that will form the project's technical stack.

Milestone: A comprehensive system design document and technology stack blueprint. This document will detail the data schema, define component APIs, and justify the selection of all key frameworks, guiding all subsequent development.

- **Phase 2: Foundation & Data Ingestion (Months 3)** This initial phase focuses on establishing the project's technical backbone and securing the data pipeline. We will set up the code repositories, server environments, and connect directly to the A2AJ API. The "Ingest" component will be built to automatically monitor, fetch, and pre-process new legal decisions and legislation as they are released.

Milestone: A stable, automated data pipeline that successfully ingests and structures new content from the A2AJ datasets, ready for AI processing.

- **Phase 3: Core AI Engine Development (Months 4-5)** This is the core research and development phase. We will build a configurable network of LLM agents described in the proposal. This includes selecting a model for legal summarization, developing the agentic workflows (using tools like *LangGraph*), and integrating the text-to-speech (TTS) or Auto podcast generation such as google NotebookLM engine for podcast generation.

Milestone: A functional proof of concept (PoC) of the AI engine that can receive a legal decision and produce an accurate, plain-language text summary and a high-quality audio podcast.

- **Phase 4: "Control Center" Interface & Testing (Months 6)** With the AI engine functional, this phase focuses on the human-centric design. We will build the simple, non-technical interface. This involves connecting the front-end controls (e.g., "Audience & Tone," "Thematic Focus") to the backend AI agent configurations.

Milestone: A functional "Control Center" that allows a non-technical user to successfully customize and generate new text and audio content.

- **Phase 5: Packaging, Deployment & Dissemination (Months 7-8)** The final phase is dedicated to packaging and releasing the platform. We will "dockerize" the entire end-to-end system into a single, easy-to-deploy kit. Comprehensive documentation for both users and developers will be written. The platform will be released as an open-source project.

Milestone: The complete, documented, and dockerized open-source platform is publicly released.

Equity, Diversity, and Inclusion

This project is fundamentally rooted in the principles of equity, diversity, and inclusion (EDI). It is designed to dismantle a primary barrier to justice: the inaccessibility of legal information for equity-deserving groups:

Applicant Experience in Fairness and Equity

The applicant has a demonstrated track record of engaging with fairness in advanced AI. This commitment is evident in current PhD research, which investigates how AI models in healthcare can discriminate against vulnerable groups based on race, gender, and age, often leading to misdiagnosis or underdiagnosis. The goal of this work is to identify and mitigate these biases to improve accuracy, fairness, and robustness in critical systems. This academic focus is substantiated by peer-reviewed publications on auditing and evaluating AI models for bias, including:

- *Fairness of AI Models in Vector Embedded Chest X-ray Representations* (NeurIPS AIM-FM, 2024)
- *Underdiagnosis Bias Mitigation With Expert Foundation Model's Representation* (IEEE Access, 2025)
- *Clinically Diverse Chest X-ray Synthesis via Cross-Modal Conditioning* (NeurIPS GenAI4Health, 2025)

This hands-on experience is directly transferable to building an equitable legal tech tool.

Project Design for Equity and Co-Creation

The "Voice of the Courts" platform directly operationalizes EDI principles by its function and design.

- The tool is explicitly designed to serve communities facing legal barriers. It empowers new Canadians by summarizing Immigration and Refugee Board decisions, supports persons with disabilities by digesting Social Security Tribunal trends, and provides access to those with literacy challenges or visual impairments through audio content.
- The project embodies the principle of "nothing about us without us" through its "Control Center" feature. This design democratizes AI, empowering legal clinics and advocacy groups to co-create the output. They are given direct control to re-task the AI with simple natural language (e.g., "Focus on refugee cases" or "Summarize changes to disability benefit rules"), making it an adaptable tool for the community.